



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2005

MATHEMATICS

8:50am – 11:55am
Monday 22nd August 2005

Directions to Students

• Reading Time : 5 minutes	• Total Marks 120
• Working Time : 3 hours	
• Write using blue or black pen. (sketches in pencil).	• Attempt Question 1 – 10
• Board approved calculators may be used	• All questions are of equal value
• A table of standard integrals is provided at the back of this paper.	
• All necessary working should be shown in every question.	
• Answer each question in the booklets provided and clearly label your name and teacher's name.	

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Total marks (120)
Attempt Questions 1 – 10
All questions are of equal value

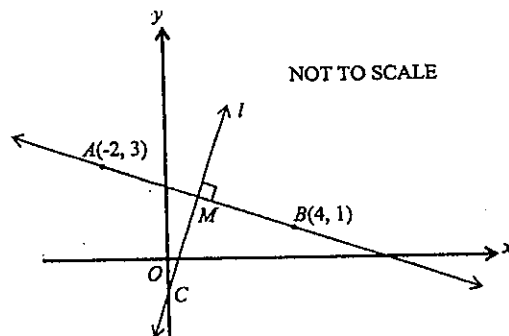
Answer each question in a SEPARATE writing booklet.

QUESTION 1 (12 MARKS) Use a SEPARATE writing Booklet Marks

- (a) The circumference of a circle is 9.29 metres. Find the radius of the circle to 3 significant figures, using the formula $C = 2\pi r$. 2
- (b) Differentiate $3x^2 - 4\sqrt{x}$. 2
- (c) Expand and simplify $(\sqrt{6} + \sqrt{2})^2$. 2
- (d) Solve the equation $a - \frac{a-2}{4} = 4$. 2
- (e) Find the values of c for which $|2c + 3| > 1$. 2
- (f) The roots of an equation are $\frac{2}{3}$ and -4 . Write down its equation in the form $ax^2 + bx + c = 0$ where a, b and c are integers. 2

QUESTION 2 (12 MARKS) Use a SEPARATE writing Booklet

Marks



The diagram shows the points $A(-2, 3)$ and $B(4, 1)$.
 M is the midpoint of the interval AB . l is the line through M perpendicular to AB .

- | | |
|---|---|
| (i) Find the gradient of the line AB . | 1 |
| (ii) Find the coordinates of M . | 1 |
| (iii) Show that the equation of AB is $x + 3y - 7 = 0$. | 1 |
| (iv) Show that the equation of l is $3x - y - 1 = 0$. | 1 |
| (v) Find the coordinates of C , the point of intersection of l and the y -axis. | 1 |
| (vi) Show that C is equidistant from A and B . | 2 |
| (vii) Find the perpendicular distance from C to the line AB . | 2 |
| (viii) Hence find the area of the triangle ABC . | 2 |
| (ix) State the inequality that defines the region of the half-plane under the line AB . | 1 |

QUESTION 3 (12 MARKS) Use a SEPARATE writing Booklet

Marks

(a) Differentiate with respect to x :

- | | |
|-----------------------|---|
| (i) $x^3 \cos x$ | 2 |
| (ii) $(3 + e^{2x})^4$ | 2 |

- | | |
|---|---|
| (b) (i) Find $\int \frac{6x}{x^2 + 2} dx$. | 2 |
| (ii) Find $\int_0^{\frac{\pi}{6}} \sec^2 2x dx$. | 2 |

(c) The gradient function of a curve is given by $\frac{dy}{dx} = 6x^2 - 4x + 1$.

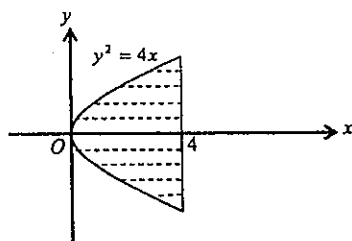
The curve passes through the point $(1, 6)$.

- | | |
|---|---|
| (i) Find the equation of the curve. | 2 |
| (ii) What is the equation of the tangent to the curve at the point $(1, 6)$? | 2 |

QUESTION 4 (12 MARKS) Use a SEPARATE writing Booklet **Marks**

- (a) Consider the function $f(x) = x^3 - 6x^2 + 9x$.
- Find the coordinates of the stationary points and determine their nature. 3
 - Find the x -coordinate of the point of inflexion. 2
 - Find where the curve meets the x -axis. 1
 - Sketch the graph of $y = f(x)$. 1
 - Find the values of x for which the gradient is decreasing. 1

(b)



The diagram shows the region bounded by the curve $y^2 = 4x$ and the line $x = 4$.

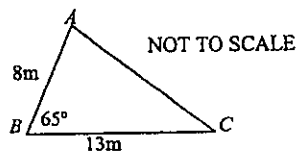
- Find the exact area of the shaded region. 2
- The region is rotated about the x -axis. Calculate the exact volume of the solid formed. 2

QUESTION 5 (12 MARKS) Use a SEPARATE writing Booklet **Marks**

- (a) Jim decides to start a daily exercise program, in which he increases the time spent on exercise each day by 2 minutes from the previous day. On the first day he does 10 minutes of exercise.
- For how long will Jim exercise on the 30th day? 1
 - What will be the total length of time spent on exercise in the first 30 days? 2
 - On which day would Jim have achieved 40 hours of exercise in total? 3
- (b) Jill commences her working career, earning \$30 000 per year in her first year. If she stays with the same company, her annual salary will increase by 5% each year.
- What will be Jill's annual salary in her 20th year (to nearest dollar)? 2
 - In what year would her annual salary first exceed \$50 000? 2
 - What will be her total earnings in the first 20 years of employment? 2

QUESTION 6 (12 MARKS) Use a SEPARATE writing Booklet **Marks**

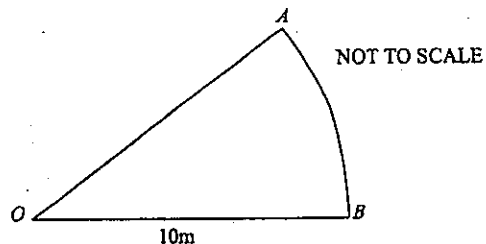
(a)



The diagram shows the triangular cross-section of the roofline of a modern building. $AB = 8\text{m}$, $BC = 13\text{m}$, and $\angle ABC = 65^\circ$. BC is horizontal.

- | | | |
|-------|--|---|
| (i) | Find the length of the beam AC . | 2 |
| (ii) | Find the angle of elevation of the beam AC . | 2 |
| (iii) | Building regulations require a cross-sectional area of at least 48m^2 for adequate ventilation. Does this design satisfy this regulation? | 1 |

(b)



AOB is the sector of a circle, centre O and radius 10m .

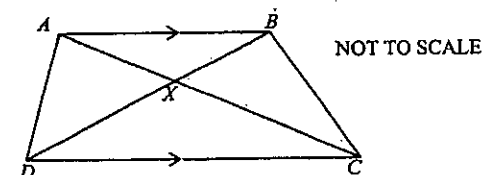
The area of the sector is 60m^2 . Calculate the exact perimeter of the sector AOB .

3

Question 6 continues on page 7

Question 6 (continued)

(c)



$ABCD$ is a trapezium, and AB is parallel to DC .
The diagonals AC and BD intersect at X . $AX = 4\text{cm}$, $CX = 10\text{cm}$, $BD = 12\text{cm}$.

Copy the diagram neatly into your answer booklet.

- | | | |
|------|---|---|
| (i) | Prove $\triangle AXB$ is similar to $\triangle CXD$. | 2 |
| (ii) | Find the length of BX . | 2 |

End of Question 6

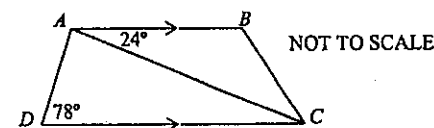
QUESTION 7 (12 MARKS) Use a SEPARATE writing Booklet Marks

- (a) Find the coordinates of the points of intersection of the line $2x - y - 2 = 0$ and the hyperbola $xy = 4$. 3
- (b) Solve the equation for x :
 $3 \times 3^{2x} + 26 \times 3^x - 9 = 0$. 3
- (c) For what values of c does the equation $x^2 + 5x + c = 0$ have two real and different roots? 2
- (d) α, β are the roots of the quadratic equation $2x^2 - 5x - 8 = 0$.
 Without finding the roots, find the value of:
- (i) $\alpha^2 + \beta^2$. 2
- (ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$. 2

QUESTION 8 (12 MARKS) Use a SEPARATE writing Booklet Marks

- (a) Evaluate $\sum_{r=2}^4 \frac{1}{r(r-1)}$. 2
- (b) The equation of a parabola is $8y = x^2 - 4x - 4$.
 By expressing the equation in the form $(x-h)^2 = 4a(y-k)$, find:
- (i) the coordinates of the vertex. 2
- (ii) the coordinates of the focus. 1
- (iii) the equation of the directrix. 1

(c)



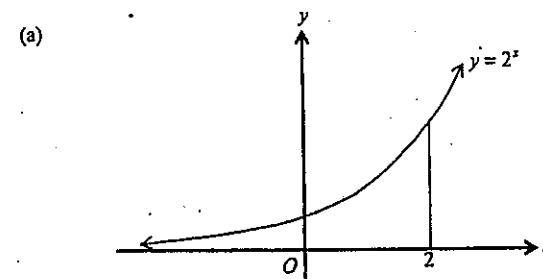
$ABCD$ is a trapezium and AB is parallel to DC . $\angle BAC = 24^\circ$, $\angle ADC = 78^\circ$. 2
 Explain why $\triangle ACD$ is isosceles.

- (d) Consider the function $f(x) = \begin{cases} 3 + 2^{-x} & \text{for } x \geq 0 \\ x + 4 & \text{for } x < 0 \end{cases}$.
- (i) Evaluate $f(-1) + f(1)$. 2
- (ii) Find $\lim_{x \rightarrow 0} f(x)$. 1
- (iii) Is the function continuous at $x = 0$? Give reasons. 1

QUESTION 9 (12 MARKS) Use a SEPARATE writing Booklet **Marks**

- (a) The value of a car depreciates exponentially according to the formula $V = Ae^{-kt}$ where V dollars is the value of the car after t years. A car's new price is \$32 000, and after 6 years its value is \$14 000.
- (i) Show that $V = Ae^{-kt}$ satisfies the equation $\frac{dV}{dt} = -kV$. 1
- (ii) What is the value of A ? 1
- (iii) Find the value of k correct to 3 significant figures. 2
- (iv) What will be the value of the car after 10 years? 1
- (v) At what rate will the value of the car be decreasing when it is 10 years old? 1
- (b) A particle moves along a straight line. Its position x metres from O , t seconds after starting, is given by $x = 3 + \cos 2t$.
- (i) Find the position of the particle after $\frac{3\pi}{4}$ seconds. 1
- (ii) In what direction does the particle start to move? 2
- (iii) Sketch the graph of v as a function of t for $0 \leq t \leq 2\pi$. 2
- (iv) What is the maximum acceleration of the particle? 1

QUESTION 10 (12 MARKS) Use a SEPARATE writing Booklet **Marks**



The area between the curve $y = 2^x$ and the x -axis for $0 \leq x \leq 2$, is rotated about the x -axis to form a solid.

- (i) Write down the definite integral which represents the volume of the solid. 1
- (ii) Use Simpson's Rule with three function values to find the approximate volume of the solid giving your answer to 2 decimal places. 2
- (iii) Using integration and the result $\int a^x dx = \frac{1}{\ln a} a^x$, evaluate the volume of the solid, giving your answer to 2 decimal places. 2
- (b) Consider the function $f(x) = xe^{-x}$.
- (i) Show that the graph of $y = f(x)$ has one stationary point at $x = 1$. 2
- (ii) Show that the stationary point is a maximum. 2
- (iii) Show that there is one point of inflexion and it occurs at $x = 2$. 1
- (iv) By considering the signs of x and e^{-x} for large positive and large negative values of x , sketch the graph of $y = f(x)$. 2

End of paper

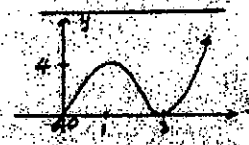
Mathematics: Question 1

Suggested Solutions	Marks Awarded	Marker's Comments
(a) $C = 2\pi r \therefore r = \frac{C}{2\pi}$ $r = \frac{9.29}{2\pi} = 1.4785 \dots$ $= 1.48 \text{ (3 s.f.)}$ (2)		
(b) $\frac{d}{dx} (3x^2 - 4\sqrt{x}) = 6x - 4 \times \frac{1}{2} x^{-\frac{1}{2}}$ $= 6x - \frac{2}{\sqrt{x}}$ (2)		
(c) $(\sqrt{6} + \sqrt{2})^2 = 6 + 2\sqrt{12} + 2$ $= 8 + 4\sqrt{3}$ (2)		
(d) $a - \frac{a-2}{4} = 4$ $4a - (a-2) = 16$ $3a + 2 = 16$ $a = 4\frac{2}{3}$ (2)		
(e) $2c+3 > 1$ $2c+3 < -1$ or $2c+3 > 1$ $2c < -4$ or $2c > -2$ $c < -2$ or $c > -1$ (2)		
(f) $(x - \frac{2}{3})(x+4) = 0$ $x^2 + 4x - \frac{2}{3}x - \frac{8}{3} = 0$ $3x^2 + 10x - 8 = 0$ (2) <u>OR</u> $(3x-2)(x+4) = 0$ $3x^2 + 10x - 8 = 0$ <u>OR</u> $x^2 - (\frac{2}{3} + 4)x + (\frac{2}{3})(-4) = 0$ $1. x^2 + 3\frac{1}{3}x - \frac{8}{3} = 0$ $1. 3x^2 + 10x - 8 = 0$		

Mathematics: Question 2

Suggested Solutions	Marks Awarded	Marker's Comments
(i) Grad. AB: $\frac{3-1}{2-4} = -\frac{1}{2}$ (1)		
(ii) M: $(\frac{-2+4}{2}, \frac{3+1}{2}) = (1, 2)$ (1)		
(iii) AB: $y-3 = -\frac{1}{2}(x+3)$ or $y-1 = -\frac{1}{2}(x-1)$ $2y-6 = -x-3$ or $2y-2 = -x+1$ $x+2y-7=0$ or $x+2y-7=0$ (1)		
(iv) L: $y-2 = 3(x-1)$ $y-2 = 3x-3$ $3x-y-1=0$ (1)		
(v) Subst. $x=0$ into L: $0-y-1=0$ $y=-1 \therefore C$ is $(0, -1)$ (1)		
(vi) $AC = \sqrt{(4-0)^2 + (2-1)^2} = \sqrt{16+1} = \sqrt{17}$ $BC = \sqrt{(4-0)^2 + (1-1)^2} = \sqrt{16+0} = 4$ $\therefore C$ is equidistant from A and B (2)		
(vii) C: $(0, -1)$ AB: $x+2y-7=0$ $d = \frac{ 0+2(-1)-7 }{\sqrt{1^2+2^2}} = \frac{9}{\sqrt{5}} = \frac{9\sqrt{5}}{5}$ (2) <u>OR</u> Dist. CM = $\sqrt{(4-0)^2 + (2-(-1))^2} = \sqrt{17}$		
(viii) $AB = \sqrt{6^2 + 2^2} = \sqrt{40}$ or $2\sqrt{10}$ Area $\triangle ABC = \frac{1}{2} \times 2\sqrt{10} \times \frac{9\sqrt{5}}{5}$ $= 18\sqrt{2}$ (2)		
(ix) Subst. $x=0, y=0$ in $x+2y-7 < 0$ LHS = $-7 < 0$ $\therefore$ Region is $x+2y-7 < 0$ (1)		

Mathematics: Question 3	Marks Awarded	Marker's Comments
Suggested Solutions (a) (i) $\frac{d}{dx} x^3 \cos x = \cos x (3x^2) + x^3 (-\sin x)$ $= 3x^2 \cos x - x^3 \sin x$ (2) (ii) $\frac{d}{dx} (3 + e^{2x})^4 = 4(3 + e^{2x})^3 \cdot 2e^{2x}$ $= 8e^{2x} (3 + e^{2x})^3$ (2) (b) (i) $\int \frac{6x}{x^2+2} dx = 3 \int \frac{2x}{x^2+2} dx$ $= 3 \log_e (x^2+2) + C$ (2) (ii) $\int_0^{\frac{\pi}{2}} \sec^2 2x dx = \left[\frac{1}{2} \tan 2x \right]_0^{\frac{\pi}{2}}$ $= \frac{1}{2} \tan \pi - \frac{1}{2} \tan 0$ $= \frac{\sqrt{2}}{2}$ (2) (c) $\frac{dy}{dx} = 6x^2 - 4x + 1$ (i) $y = 2x^3 - 2x^2 + x + C$ $x=1, y=6: 6 = 2 - 2 + 1 + C$ $C = 5$ Curve is $y = 2x^3 - 2x^2 + x + 5$ (2) (ii) When $x=1, \frac{dy}{dx} = 6 - 4 + 1$ $= 3$ Egn. of tangent is $y - 6 = 3(x - 1)$ (2) i.e. $3x - y + 3 = 0$ or $y = 3x + 3$		

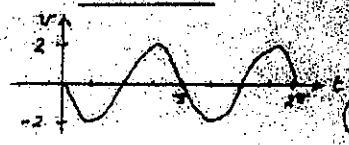
Mathematics: Question 4	Marks Awarded	Marker's Comments
Suggested Solutions (a) $f(x) = x^3 - 6x^2 + 9x$ (i) $f'(x) = 3x^2 - 12x + 9$ $= 3(x^2 - 4x + 3)$ $= 3(x-1)(x-3) = 0$ when $x=1, 3$ $f''(x) = 6x - 12$ $f(1) = 1 - 6 + 9 = 4; f''(1) = 6 - 12 < 0$ $\therefore$ Maximum at $(1, 4)$ $f(3) = 27 - 54 + 27 = 0; f''(3) = 18 - 12 > 0$ $\therefore$ Minimum at $(3, 0)$ (3) (ii) Pt. of inflexion: $6x - 12 = 0$ $x = 2$ If $x < 2, f''(x) < 0$; If $x > 2, f''(x) > 0$ Change in concavity pt. of inflexion at $x = 2$ (2) (iii) When $y=0, x^3 - 6x^2 + 9x = 0$ $3x(x-3)^2 = 0$ Meets x-axis at $0, 3$ (1) (iv)  (1) (v) Grad. decreasing for $x < 2$. (1) (b) (i) $A = 2 \int_0^4 x x^{\frac{1}{2}} dx = 4 \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^4$ $= \frac{8}{3} [4^{\frac{3}{2}} - 0]$ Area = $21\frac{1}{3}$ unit² (2) (ii) $V = \pi \int_0^4 4x dx = \pi [2x^2]_0^4$ $= 2\pi [4^2 - 0]$ Volume = 32π unit³ (2)		

Mathematics: Question 5		
Suggested Solutions	Marks Awarded	Marker's Comments
(a) 10, 12, 14, ... (i) $T_{30} = a + (n-1)d = 10 + 24 \times 2$ Time = 68 minutes (1) (ii) $S_{30} = \frac{a}{2}(a + l)$ or $\frac{a}{2}[2a + (n-1)d]$ $= \frac{10}{2}(10 + 68)$ Total time = 1170 minutes (2) (iii) 40 hours = 2400 minutes. $\frac{a}{2}[2a + (n-1)d] = 2400$ $\frac{a}{2}[20 + (n-1)2] = 2400$ $n(10 + n - 1) = 2400$ $9n + n^2 - 2400 = 0$ $n = \frac{-9 \pm \sqrt{81 - 4(-2400)}}{2}$ $= 44.7$ (since $n > 0$) 40 hours reached on 45th day (3) (b)(i) $T_{20} = ar^{n-1} = 30000 \times (1.05)^{19}$ Salary = \$75809 (2) (ii) $30000 \times 1.05^{n-1} > 50000$ $1.05^{n-1} > 1\frac{1}{3}$ $(n-1) \log 1.05 > \log 1\frac{1}{3}$ $n-1 > \frac{\log 1\frac{1}{3}}{\log 1.05}$ $n > 11.47$ (2) Exceeds \$50000 in 12th year (iii) $S_{20} = \frac{a(r^n - 1)}{r - 1} = \frac{30000(1.05^{20} - 1)}{1.05 - 1}$ Total salary = \$991979 (2)		

Mathematics: Question 6		
Suggested Solutions	Marks Awarded	Marker's Comments
(a)(i) $AC^2 = 8^2 + 13^2 - 2 \times 8 \times 13 \times \cos 65^\circ$ $AC = 12.05 \text{ m.}$ (2) (ii) $\frac{\sin \angle ACB}{8} = \frac{\sin 65^\circ}{12.05}$ (2) $\sin \angle ACB = \frac{8 \sin 65^\circ}{12.05} = 0.6017$ $\angle ACB = 37^\circ$ (nearest degree) (iii) Area = $\frac{1}{2} \times 8 \times 13 \times \sin 65^\circ$ $= 47.12 \text{ m}^2$ Does not meet regulation. (1) (b) $A = \frac{1}{2} \pi \times 10 \therefore 60 = \frac{1}{2} \times 10^2 \times \theta$ $\theta = 1.2$ Perimeter = $10 + 10 + 10 \times 1.2$ $= 32 \text{ m.}$ (3) (c) (i) In $\triangle AEB$, $\triangle CED$, $\begin{cases} \angle AEB = \angle CED \text{ (vertically opposite)} \\ \angle EAB = \angle EDC \text{ (alternate, AB DC)} \\ \angle EBA = \angle ECD \text{ (alternate, AD BC)} \end{cases}$ $\therefore \triangle AEB \cong \triangle CED$ (2 angles equal) (2) (ii) Let $BE = x \text{ cm}$, then $DE = (3-x) \text{ cm}$ $\frac{4}{x} = \frac{10}{3-x}$ $4(3-x) = 10x$ $12 - 4x = 10x$ $12 = 14x$ $x = \frac{12}{14}$ $BE = \frac{6}{7} \text{ cm (or equivalent)}$ (2)		

Mathematics Question 7		Marks Awarded	Marker's Comments
(a) $2x - y - 2 = 0$ or $y = 2x - 2$ $2x - y = 4$ $x(2x - 2) = 4$ $2x^2 - 2x - 4 = 0$ $x^2 - x - 2 = 0$ $(x - 2)(x + 1) = 0$ $x = 2, -1$ (3) Points of intersection: $(2, 2), (-1, -4)$			
(b) $3 \times 3^{2x} + 26 \times 3^x - 9 = 0$ Let $u = 3^x$, $3u^2 + 26u - 9 = 0$ $(3u - 1)(u + 9) = 0$ $u = \frac{1}{3}$ or $u = -9$ $3^x = \frac{1}{3}$ or $3^x = -9$ $x = -1$ (no solution) (3)			
(c) $x^2 + 5x + 6 = 0$ For real and different roots, $a > 0$ $25 - 4c > 0$ $c < 6\frac{1}{4}$ (2)			
(d) $2x^2 - 5x - 8 = 0$ $\alpha + \beta = \frac{5}{2}$, $\alpha\beta = \frac{-8}{2} = -4$ (1) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $= \frac{25}{4} - 2(-4)$ $= 14\frac{1}{4}$ (2)			
(ii) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$ $= \frac{14\frac{1}{4}}{-4}$ $= -\frac{57}{16}$ (2)			

Mathematics Question 8		Marks Awarded	Marker's Comments
(a) $\sum_{r=2}^n \frac{1}{r(r-1)} = \frac{1}{2 \times 1} + \frac{1}{3 \times 2} + \frac{1}{4 \times 3}$ (2) $= \frac{3}{4}$			
(b) $2y = x^2 - 4x + 4$ $x^2 - 4x + 4 = 2y + 4 + 4$ $(x - 2)^2 = 2(y + 4)$ NB $a = 2$. (i) Vertex $(2, -4)$ (2) (ii) Focus $(2, -1)$ (1) (iii) Directrix $y = -5$ (1)			
(c)  $\angle ACD = \angle BAC$ (alternate, $AB \parallel DC$) $= 24^\circ$ $\angle DAC = 180^\circ - (78^\circ + 24^\circ)$ (angle sum of $\triangle ACD$) $= 78^\circ$ $\therefore \angle DAC = \angle ADC$ $\triangle ACD$ is isosceles (2)			
(d) $f(x) = \begin{cases} 3 + 2^{-x} & \text{for } x < 0 \\ x + 4 & \text{for } x \geq 0 \end{cases}$ (i) $f(-1) + f(1) = (-1 + 4) + (3 + 2^{-1})$ $= 3 + 3 + \frac{1}{2}$ $= 6\frac{1}{2}$ (2)			
(ii) $\lim_{x \rightarrow 0} f(x) = 3 + 0 = 3$ (1)			
(iii) As $x \rightarrow 0$ from below, $f(x) \rightarrow 3 + 2^{-x} = 4$ As $x \rightarrow 0$ from above, $f(x) \rightarrow 3 + 2^{-x} = 4$ $\therefore f(x)$ is continuous at $x = 0$. (1)			

Mathematics: Question 9		Marks Awarded	Marker's Comments
Suggested Solutions			
(a) $V = Ae^{-kt}$			
(i) $\frac{dV}{dt} = -kAe^{-kt}$ $= -kV$	(1)		
(ii) When $t=0$, $V=32\,000$ $32\,000 = Ae^0$ $A = 32\,000$	(1)		
(iii) $t=6$, $V=14\,000$: $14\,000 = 32\,000e^{-6k}$ $e^{-6k} = \frac{14\,000}{32\,000}$ $-6k = \log_e\left(\frac{14}{32}\right)$ $k = 0.102 \text{ (3 s.f.)}$	(2)		
(iv) When $t=10$, $V = 32\,000 \times e^{-0.102 \times 10}$ Value = \$8051	(1)		
(v) When $t=10$, $\frac{dV}{dt} = -0.102 \times 8051$ $= -821$ Value decreasing at \$821 per year	(1)		
(b) $x = 3 + 10s \sin t$			
(i) When $t = \frac{3\pi}{2}$, $x = 3 + 10 \cos\left(2 \times \frac{3\pi}{2}\right)$ $= 3$	(1)		
(ii) $v = -2 \sin 2t$ It starts to move in negative direction	(2)		
(iii)	(2)		
			
(iv) $a = -4 \cos 2t$ Maximum acceleration is 4 m/s^2	(1)		

Mathematics: Question 10		Marks Awarded	Marker's Comments
Suggested Solutions			
(a) (i) $V = \pi \int_0^2 (2^x)^2 dx$		(1)	
(ii)	(2)		
$\int_0^2 (2^x)^2 dx = \frac{1}{\ln 4} [1 + 4 \times 4 + 16]$ $= 11$ Volume = $11 \times \pi = 34.56 \text{ unit}^3 \text{ (2 d.p.)}$			
(iii) $V = \pi \int_0^2 4^x dx$ $= \pi \left[\frac{1}{\ln 4} 4^x \right]_0^2$ $= \frac{\pi}{\ln 4} [16 - 1]$ Volume = $33.99 \text{ unit}^3 \text{ (2 d.p.)}$		(2)	
(b) $f(x) = xe^{-x}$			
(i) $f'(x) = e^{-x} + x(-e^{-x})$ $= e^{-x}(1-x)$ $= 0$ only when $x=1$ $\therefore$ only one stationary point	(2)		
(ii) $f''(x) = (1-x)(-e^{-x}) + e^{-x}(-1)$ $= xe^{-x} - 2e^{-x} = e^{-x}(x-2)$ $f''(1) = 1e^{-1} - 2e^{-1} < 0$ $\therefore$ Stat. point is maximum at $x=1$ OR show derivative changes sign	(2)		
(iii) For $f'(x) = 0$, $x=2$ $\therefore x=2$ is the only point of inflection	(1)		
(iv) For $x > 0$, $e^{-x} > 0 \therefore f(x) > 0$ For $x < 0$, $e^{-x} > 0 \therefore f(x) < 0$	(2)		
